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Project report - Wind Engineering

Cross wind on road vehicles

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1. Introduction and Background

Road freight transport plays a crucial role in the economy, accounting for a significant portion of goods movement across Europe. However, each year, over 4.000 fatalities in the EU are linked to accidents involving heavy goods vehicles, with rollovers due to strong crosswinds being a notable contributing factor, particularly on exposed roads such as bridges and open highways.

Accurate modeling of crosswind effects can enhance safety by predicting critical scenarios, thus enabling improvements on the design phase of the vehicle and highliting what precautions are needed for each situation.

This study focuses on the dynamic response of a truck subjected to different lateral wind profiles, with different velocities and turbulences. The model integrates key vehicle parameters, such as mass distribution, geometry, and aerodynamic coefficients, alongside a driver model as a PDI controller. Simulations are carried out under varying road and load conditions, with particular attention to rollover risk. Additionally, lateral trajectory deviation is considered, as even modest lane departure can pose a serious threat in real-world traffic.

2. Objectives

The rollover behavior of the truck will be characterized by the minimum velocity at which rollover occurs and by the lateral lane deviation, quantified through its mean value over the entire trajectory. For the first two parts of the study, the trajectory will be assumed to be a straight line.

First, simulations will be carried out using various wind profiles — corresponding to intensities of 25 and 30 m/s, and turbulence intensities of 7%, 14% and 25% under different road conditions: dry, wet, and snowy surfaces. The goal is to identify the general trends and sensitivities of the vehicle's response to these environmental factors.

Next, the influence of vehicle dynamics on rollover risk will be investigated by varying the payload distribution (λ) and the roll stiffness balance between axles (τ_{roll}).

Finally, non-linear trajectories such as turns and lane changes will be studied to assess their impact on both rollover potential and lateral deviation.

3. Methodology

In this study, wind is always assumed to act along the absolute Y-axis direction. Thus, with V being the vehicle speed, the relative wind velocity is:

$$\overline{V}_{\text{rel}} = \overline{U} - \overline{V} \tag{1}$$

As shown in Figure 1, the angle between $\overline{V}_{\text{rel}}$ and the truck's longitudinal axis defines the angle of attack α, which is used to compute the aerodynamic coefficients. These coefficients were obtained through wind tunnel testing using a 1:10 scale model of the truck and are assumed to depend only on α. The lateral force, roll moment, and yaw moment are respectively expressed as:

$$F_{yw} = \frac{1}{2}\rho|\overline{v}_{\text{rel}}|^2 Alc_y(\alpha) \tag{2}$$

$$M_{xw} = \frac{1}{2}\rho|\overline{v}_{\text{rel}}|^2 Alh_r c_{mx}(\alpha) \tag{3}$$

$$M_{zw} = \frac{1}{2}\rho|\overline{v}_{\text{rel}}|^2 Alh_r c_{mz}(\alpha) \tag{4}$$

It is important to note that the same height h_r and area A are used as the characteristic length and surface, respectively, for both moment expressions along the x and z axes.

The dynamic system of the truck, considering v_y, ρ, and $\dot{\psi}$ as degrees of freedom, is given by:

$$\begin{cases} ma_y = F_{yF} + F_{yR} + F_{yW} \\ J_z\ddot{\psi} = F_{yF}a - F_{yR}b + M_{zW} + F_{yW}x_P \\ J_x\ddot{\rho} = m(z_G - z_R)(a_y + g\sin\rho) - M_A - M_P + M_{xW} - F_{yW}(z_P - z_R) \end{cases} \tag{5}$$

The lateral tire forces (F_{yF} and F_{yR}) are computed using the Pacejka model, with the normal loads depending on the value of τ_{roll}.

Solving this system yields the vertical reaction force at each tire over time. Rollover is considered to occur when any of the vertical forces drops below 10% of its respective static value, a conservative threshold adopted instead of the actual zero-load condition.

It is important to emphasize that the chosen time step Δt affects both the convergence and validity of the simulation; hence, it is adapted in each case to ensure convergence.

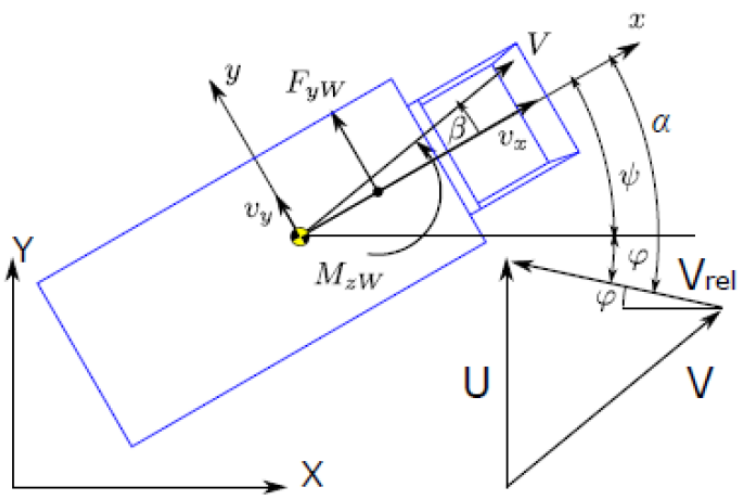


Figure 1: Absolute and truck reference frames.

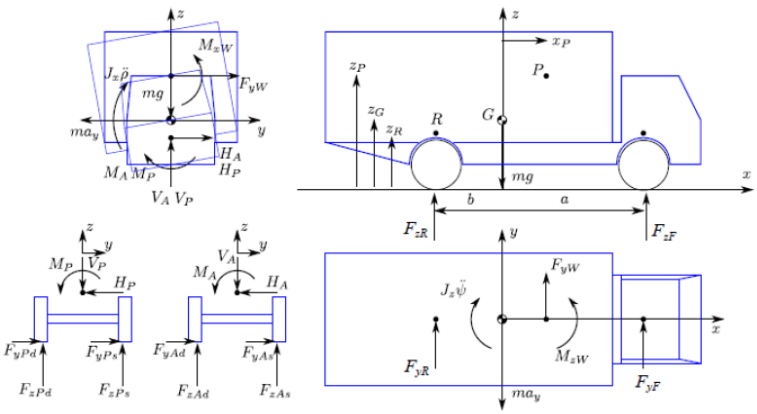


Figure 2: Dynamic equilibrium.

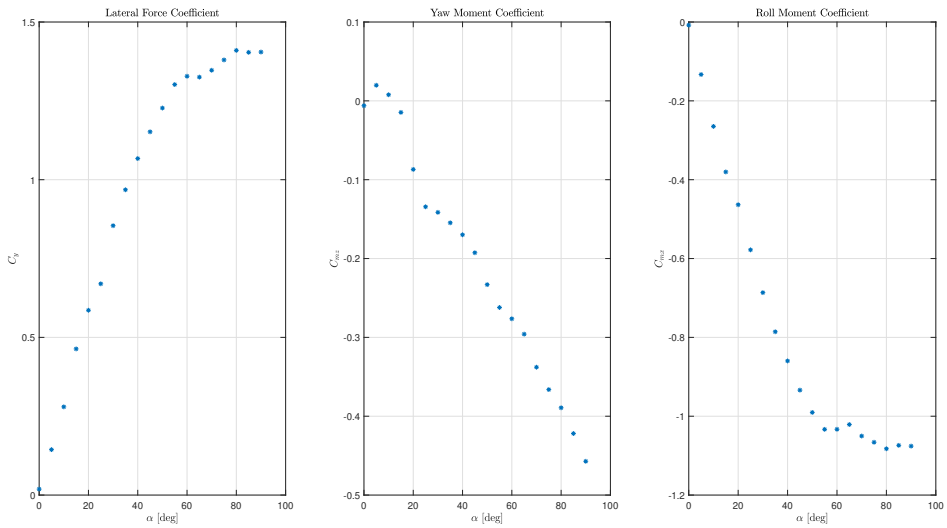


Figure 3: Aerodynamic coefficients as function of the angle of attack.

4. Results

4.1. Characteristic Wind Curves and Wind Field Analysis

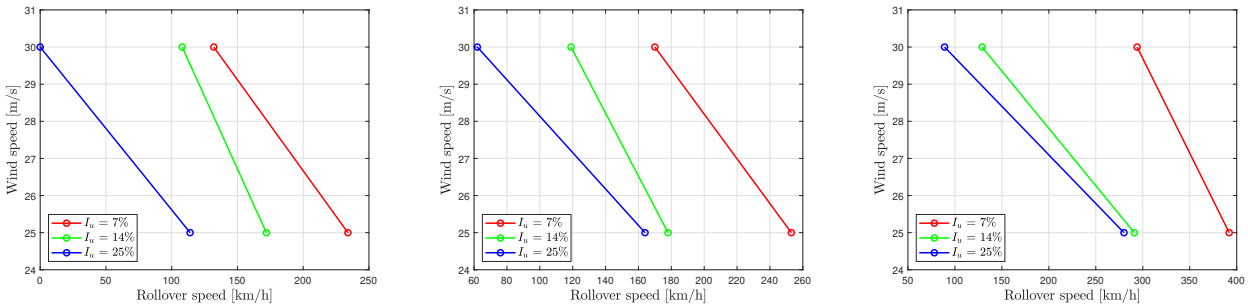
One of the main objectives of analyzing the interaction between crosswind and road vehicles is to determine the **Characteristic Wind Curves (CWC)**. These curves represent, for a given wind condition, the maximum vehicle speed at which the truck can operate safely without experiencing a rollover.

To construct the CWC, simulations were carried out under different wind scenarios by varying the mean wind speed and turbulence intensity. The main wind parameters considered were:

- **Mean wind speed**: 25 m/s and 30 m/s
- **Turbulence intensity**: 7%, 14%, and 25%

A total of six wind histories were generated from these combinations. For each case, the truck speed was gradually increased until a rollover event occurred. The simulations were constrained by the physical limits of the wind field, which was available over a time duration of 100 s and a spatial domain of 1000 m. Consequently, the final simulation time was set as:

$$t_{\text{final}} = \min\left(t_{\max}, \frac{x_{\max}}{v_{\text{truck}}}\right)$$



(a) Dry. (b) Wet. (c) Snow.

Figure 4: Characteristic wind curve.

The CWC results are presented in Fig. 4 for three road surface conditions: *dry*, *wet*, and *snow*. As expected, for each surface condition, higher wind turbulence or stronger average wind speeds lead to lower rollover thresholds.

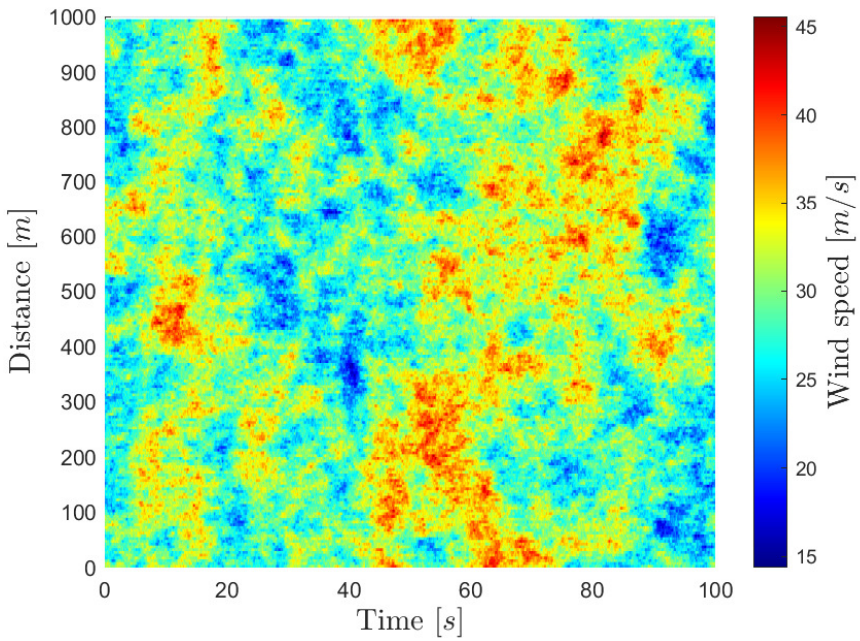


Figure 5: Wind field for a cross wind of 30 m/s and a turbulent index of 14%

In addition, the wind field corresponding to the case of 30 m/s mean wind speed and 14% turbulence intensity is visualized in Fig. 5. This map reveals how wind gusts are distributed over space and time. Strong localized peaks in wind speed can be observed, which are critical in triggering rollover, especially at low vehicle velocities where the wind dominates the relative airflow.

A particularly dangerous region is observed between 40 and 60 seconds within the first 400 m of the domain, where wind gusts exceed 40 m/s. This intense burst of wind is especially relevant for low-speed conditions, as vehicles remain longer in this region and are more exposed to the gusts. Such localized wind events can induce rollover even when the vehicle is nearly stationary, highlighting the importance of accounting for transient and spatially varying wind effects when assessing safety thresholds.

These findings emphasize the importance of considering both road and wind conditions when defining safe operational envelopes for heavy vehicles. The CWC provide a clear graphical tool to identify safe vehicle speeds under varying environmental conditions.

4.2. Effect of road conditions on rollover behavior

In this section, the analysis under the three different road conditions (dry, wet, snow) is conducted, where each scenario have a different friction coefficient. The resulting rollover speeds are summarized in Table 1.

An interesting observation is that rollover speed increases as road friction decreases. Although this may seem counter-intuitive at first, it becomes clear when considering the underlying physics: with lower friction, the tyres provide less lateral grip, making it harder for them to act as a pivot during rollover. In many of the wet and snowy scenarios, the rollover speeds become so high that they exceed the truck's physical capabilities, making an actual rollover practically unachievable in those conditions.

Although the rollover condition is delayed under low-friction conditions such as wet or snow-covered roads, these scenarios remain highly dangerous. The reduced tyre grip significantly increases lateral displacement, making it harder for the driver to maintain control of the vehicle. As a result, even if the rollover threshold is not reached, the truck may experience large deviations from its intended path or become unstable, posing serious safety risks in real driving situations.

Table 1: Rollover speed $[km/h]$ in different road conditions

	Turbulence index								
	7%			14%			25%		
Wind Speed $[m/s]$	Dry	Wet	Snow	Dry	Wet	Snow	Dry	Wet	Snow
25	234	253	392	172	178	291	114	164	280
30	132	170	294	108	119	129	0	62	89

4.3. Effect of payload factor (λ) on rollover and centre of Gravity Displacement

The effect of the payload, represented by the parameter λ, is evaluated by varying it in the range $[0, 1]$, where $\lambda = 1$ corresponds to the vehicle carrying its maximum admissible payload. This parameter influences the vertical position of the vehicle's center of mass and consequently affects its susceptibility to rollover.

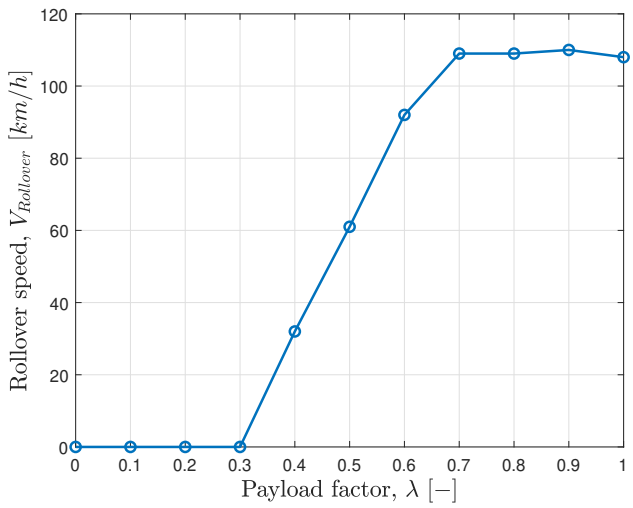
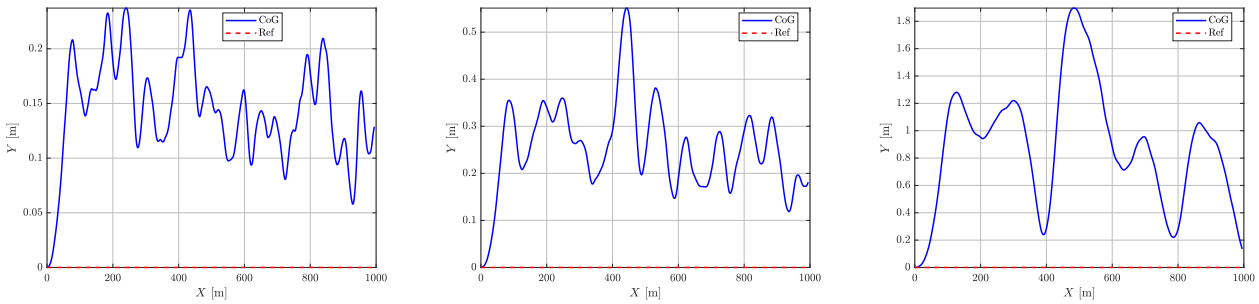


Figure 6: Rollover speed as function of payload in dry road condition for a crosswind speed of 30 m/s, a turbulence intensity of 14%, and a ratio of the roll stiffness $\tau_{Roll} = 0.664$

In Fig. 6, the rollover velocity is analyzed as a function of λ. For values of λ between 0 and 0.3, the rollover velocity is zero, indicating complete instability. This behavior can be attributed to the high position of the center of gravity at low λ values, which produces greater moments from the tyres and increases the risk of rollover. As λ increases, the center of gravity lowers, and the rollover velocity also increases until it reaches a plateau, remaining nearly constant between $\lambda = 0.7$ and $\lambda = 1$.



(a) Dry. (b) Wet. (c) Snow.

Figure 7: Displacement of centre of gravity of the truck in the three road conditions for a cross wind of 30 m/s and a turbulent index of 14%.

In addition, in the Fig. 7 the displacement of the center of gravity of the truck is evaluated as a function of the vehicle's longitudinal displacement x under three different road conditions: dry, wet, and snowy. All simulations are conducted for a crosswind speed of 30 m/s and a turbulence intensity of 14%.

The results show that, under dry road conditions, the displacement of the center of gravity exhibits numerous sharp peaks. This is due to the strong grip of the tires, which results in abrupt vehicle responses to aerodynamic disturbances. In contrast, under snowy conditions, the displacement shows fewer oscillations, but they are of larger amplitude. This is due to the lower tire-road friction, which reduces control authority and allows larger deviations in the vehicle trajectory.

4.4. Effect of roll stiffness distribution on rollover behavior

In this section, the influence of the load transfer distribution between axles is evaluated by modifying the parameter τ_{roll}, which represents the ratio of the roll stiffness of the front axle relative to the total roll stiffness of the vehicle. This parameter affects the distribution of the load transfer but not its total amount. The analysis is performed for a crosswind speed of 30 m/s, a turbulence intensity of 14%, in dry road condition and a relative payload factor $\lambda = 1$.

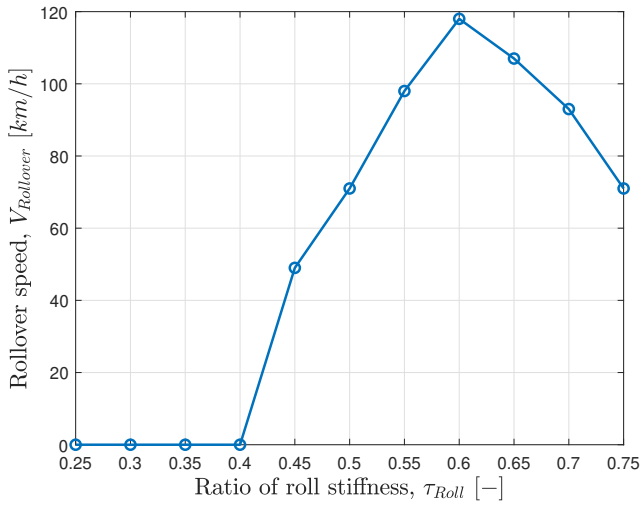
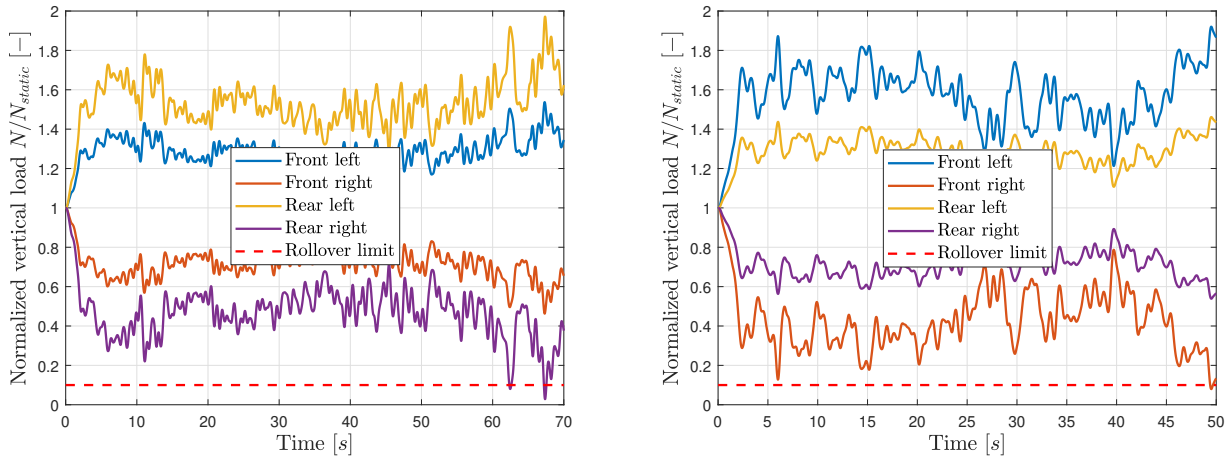


Figure 8: Rollover speed as function of the ratio of roll stiffness in dry road condition for a crosswind speed of 30 m/s, a turbulence intensity of 14%, and a relative payload factor $\lambda = 1$

First, the rollover velocity is computed as a function of τ_{roll}. The resulting plot (Fig. 8) shows that the maximum rollover velocity is reached at $\tau_{\text{roll}} = 0.6$. Furthermore, for values of τ_{roll} between 0.25 and 0.4, the rollover velocity is zero. This indicates that the vehicle is completely unstable under these conditions and would rollover even at zero forward speed.



(a) $\tau_{Rollover} = 0.45$.

(b) $\tau_{Rollover} = 0.75$.

Figure 9: Load on each wheel as a function of time in dry road condition for a crosswind speed of 30 m/s, a turbulence intensity of 14%, and a relative payload factor $\lambda = 1$

Secondly, the Fig. 9 shows the normalized vertical wheel loads are analyzed for two specific values of τ_{roll}: 0.45 and 0.75. At $\tau_{\text{roll}} = 0.45$, the rear right wheel is the first to lift off, leading to rollover. At $\tau_{\text{roll}} = 0.75$, the front right wheel is the one that lifts off instead. This behavior is consistent with the fact that the wind is coming from the right side, making it logical that the right-hand wheels are the first to lose contact with the ground.

4.5. Influence of Turning Maneuvers on Rollover Speed

As follows, the effect of a turn on the rollover speed is the final parameter to be analyzed, in comparison to the straight-line scenario considered so far. For this case, the driver is modeled as a *path follower* using a PD controller. Given a reference trajectory as input, the controller computes the required steering angle (δ) for the front wheels at each time step. This angle is determined according to the following control law:

$$\delta = k_p \left(Y_{\text{ref}} - Y_{\text{oss}}\right) + k_d \left(\dot{Y}_{\text{ref}} - \dot{Y}_{\text{oss}}\right) \tag{6}$$

where k_p and k_d are the proportional and derivative gains, respectively, and they multiply the difference between the reference trajectory and the observed one (with an observation distance set to 3 m), or the difference between their derivatives. In the case of the reference trajectory, its derivative was estimated using the finite difference method.

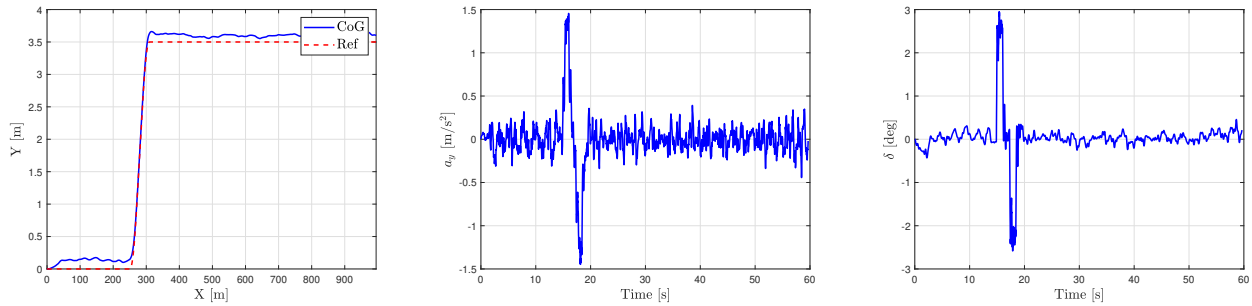
4.5.1 Single lane change

The analysis was performed by assigning the driver a reference trajectory corresponding to a lane change (assuming a lane width of 3.5 m), to be completed within a 40 m distance, first turning left, then right. In this way, the same maneuver is carried out under identical wind conditions ($V = 30m/s, I_u = 14\%, L_u = 150m$), with the only difference being the direction of the turn.

In dry road conditions, the truck overturns at 69 km/h when turning left and at 76 km/h when turning right. This asymmetry may appear counterintuitive: with wind coming from the right, one might expect that turning right, where both the centrifugal force and the wind force act to the left, would result in a lower rollover velocity. However, the simulation shows the opposite. The reason for this is the correction maneuver performed after the truck reaches the desired lateral displacement. This maneuver essentially involves a right turn, and it is at this point that the rollover occurs in the simulations. The difference in rollover speeds is due to the larger steering angle required for the left lane change, which results in a higher centrifugal force.

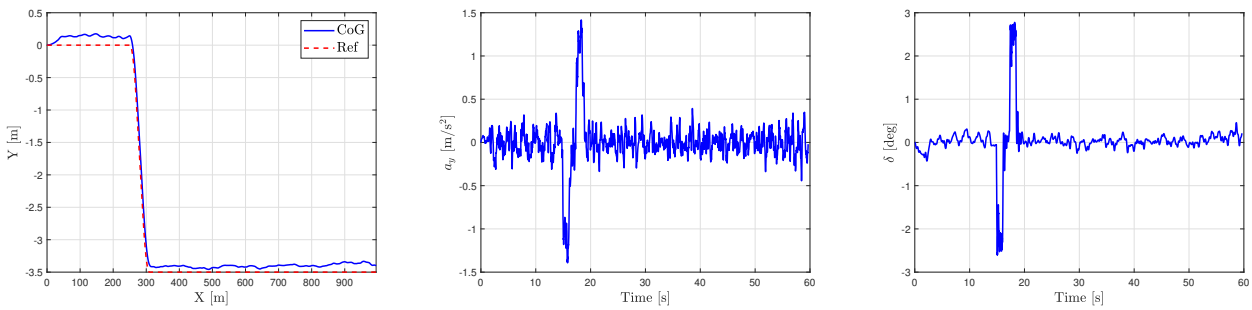
Figs. 10 and 11 show the trajectory, steering angle δ, and lateral acceleration a_y for left and right turns at $60km/h$ (a safe velocity to avoid rollover) for a dry road condition. It is evident that both δ and a_y exhibit

oscillations during the turn, but in opposite directions. Moreover, the driver is able to follow the curve with high precision, maintaining a deviation from the reference trajectory of less than 30 cm throughout the maneuver.



(a) Trajectory. (b) Lateral acceleration. (c) Steering angle.

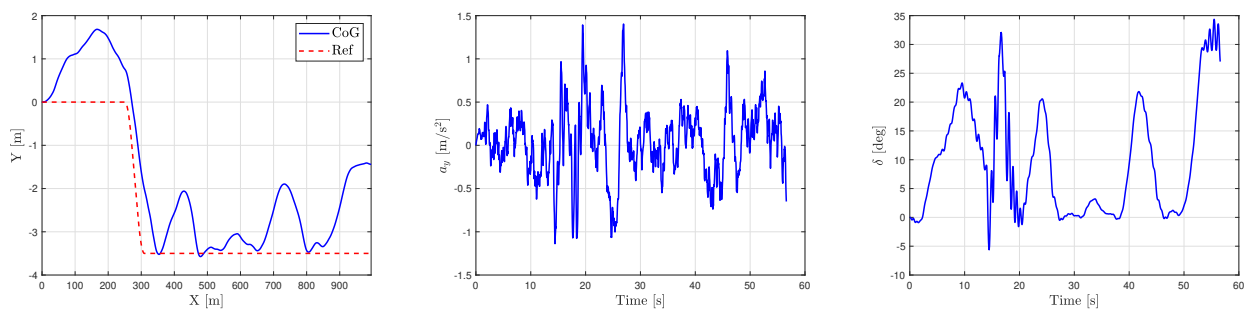
Figure 10: Left turn - Dry road - Wind ($V = 30m/s$, $I_u = 14\%, L_u = 150m$)



(a) Trajectory. (b) Lateral acceleration. (c) Steering angle.

Figure 11: Right turn - Dry road - Wind ($V = 30m/s$, $I_u = 14\%, L_u = 150m$)

Snow road condition was also considered to assess how a slippery surface would affect trajectory tracking. In particular, Fig.12 reports the trajectory, the lateral acceleration and the steering angle for a left turn at 60 km/h on the snow condition. It is evident that the lack of grip makes trajectory tracking more difficult, requiring much higher steering angles, reaching high values of steering (inside the limit of 45°).



(a) Trajectory. (b) Lateral acceleration. (c) Steering angle.

Figure 12: Right turn - Snow road - Wind ($V = 30m/s$, $I_u = 14\%, L_u = 150m$)

Finally, a representation of the variation of the rollover velocity as a function of the payload fraction was computed for three cases: *straight*, *left turn*, and *right turn*. In Fig. 13, it can be seen that the rollover speed increases with payload up to around $\lambda = 0.7 - 0.8$, after which it slightly decreases. This behavior results from

two competing mechanisms. On one hand, increasing payload improves stability by lowering sensitivity to wind disturbances and improving tire grip due to higher normal forces. On the other hand, although excessive payload lowers the center of gravity, its increased weight amplifies the centrifugal load, which ultimately reduces the rollover threshold. The straight-line case exhibits the highest rollover speeds throughout, while turning results in lower thresholds due to additional lateral acceleration and adverse wind interaction.

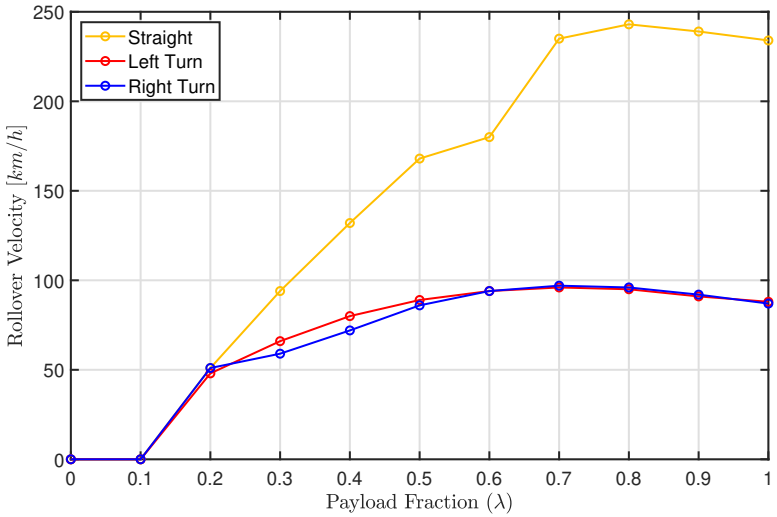


Figure 13: Rollover speed as function of payload in dry road condition and with a cross wind of 25 m/s and a turbulent index of 7%

4.5.2 Double lane change (overtaking)

To further investigate the interaction between steering maneuvers and crosswind effects, a second test was conducted involving an overtaking maneuver. In this case, the truck performs a double lane change: first turning left to overtake, then right to return to its original lane, mimicking a realistic highway scenario. This can be seen in Fig.14

The reference trajectory consists of two successive lateral displacements of $3.5m$ each, with the total maneuver completed over a longitudinal distance of $580m$: $40m$ for the initial lane change, $500m$ driving in the adjacent lane, and $40m$ for the return to the original lane. The same wind conditions are maintained as in the previous single-lane-change analysis ($V = 30\,\mathrm{m/s}$, $I_u = 14\%$, $L_u = 150\,\mathrm{m}$).

Again, the truck speed is set to $60km/h$, which remains below the rollover limits previously identified for single lane changes under dry road conditions. In this scenario, the effective rollover limit is determined by the lower of the two turning thresholds (left or right) and in this case, it corresponds to the left turn, as explained before.

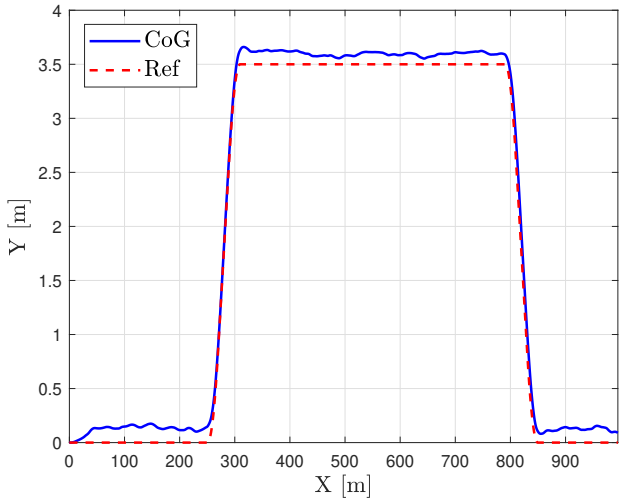


Figure 14: Trajectory of the truck following a double lane change (overtaking)

5. Conclusions

This work has presented a comprehensive numerical investigation on the rollover behavior of heavy road vehicles under crosswind conditions. The study emphasizes the critical influence of environmental and vehicle parameters (such as wind intensity, road surface condition, payload distribution, roll stiffness ratio, and turning maneuvers) on the dynamic stability of trucks.

The construction of **Characteristic Wind Curves (CWC)** under various combinations of mean wind speed and turbulence intensity proved to be an effective method to define operational safety thresholds. These curves clearly show that increasing wind severity leads to a reduction in rollover velocity, especially under dry road conditions where tyre grip is highest.

Interestingly, the analysis revealed that **low-friction surfaces such as wet or snow-covered roads can delay the onset of rollover**, not due to increased stability but due to a lack of grip that prevents the tyres from acting as a pivot point. However, this does not translate into safer driving conditions: large lateral displacements and poor control authority in these scenarios can still pose severe safety risks.

The effect of **payload distribution** showed a non-linear behavior. At low payloads, the vehicle becomes unstable, and its dynamic behavior improves with its weight. A similar non-monotonic response was found in the **roll stiffness distribution**, where the rollover speed reached a maximum at intermediate values of the front-to-total roll stiffness ratio. This underscores the importance of carefully tuning suspension characteristics in truck design.

The study also confirmed that **vehicle maneuvering**, such as turning under crosswind exposure, significantly reduces the rollover threshold. The inclusion of a steering controller allowed for a realistic assessment of this behavior, demonstrating that combined crosswind and path-following scenarios should not be overlooked in safety analysis.

Overall, the results demonstrate that **crosswind-induced rollover is a highly multidimensional problem** that cannot be evaluated based solely on wind speed or vehicle speed. A thorough understanding of road conditions, aerodynamic loading, and internal vehicle dynamics is essential. The methodology and findings presented in this work offer a valuable framework for future studies and for developing active safety systems in heavy vehicles.